Econ 5725

Quantitative Macro

Wednesdays 4pm-7pm



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Course Description: This class follows the first year PhD macro sequence, Econ 501 and 502, and the class on advanced quantitative methods in economics, Econ 5121. The goal of this course is to (1) introduce the students to modern quantitative macroeconomics, and (2) prepare students to solve recent quantitative model in macroeconomics. Applications include models with rich heterogeneity across agents and firms, aggregate fluctuations, and transitional dynamics. This course will require intensive computational work by students. Although this is not a programming class, some knowledge of computer programming is necessary. If you do not know how to program you will need to work harder. You are free to use whatever programming language you feel comfortable with.

Text(s): We will use several lecture notes available online. However, a good reference is the book Dynamic "General Equilibrium Modeling: Computational Methods and Applications," by Burkhard Heer and Alfred Maussner.

Grade Distribution: Your grade will be based on your grades in the homework assignments and a short final project.

Course Policies:

- **Labs and Assignments**
  - Students are expected to work independently. **Offering** and **accepting** solutions from others is an act of **plagiarism**, which is a serious offense and **all involved parties will be penalized according to the Academic Honesty Policy**. Discussion amongst students is encouraged, but when in doubt, direct your questions to the professor, tutor, or lab assistant.
  - **No late assignments will be accepted under any circumstances**.
- **Attendance and Absences**
  - Attendance is expected at each class.
  - Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Week	Content
Week 1	• **Why we do quantitative macroeconomics?** – Lucas (1980), Kydland-Prescott (1996). – **Deterministic Growth Model**. – Extended Path Method. – King and Rebelo (1993).
Week 2	• **Stochastic Growth Model**. – Discrete-State-Space Dynamic Programming Problem. – Value function iteration with discretization. – Howards improvement. Other ways to speed-up the algorithm – **Business Cycles**. – Greenwood, Hercowitz, and Huffman (1988). – Computing the model's statistics
Week 3	• **Households Heterogeneity in Macroeconomics**. – Why? Krueger, Mitman, and Perri (new handbook chapter). – Macro-micro facts – Kaplan and Violante (2010), Guvenen, Ozkan, and Song (2014) – Data sets. SFC. Kuhn and Rios-Rull (2005).
Week 4	• **The income-fluctuation problem**. – Policy function iteration. 5 alternative methods. – Non-iterative methods. – Accuracy.
Week 5	• **Stationary Distribution and GE**. – Monte Carlo and the invariant density function. – Aiyagari. Solving for r^*.
Week 6	• **No class.**

Week 7	• **Life-cyle models of consumption, debt, and default**. – Life-cycle models. – Unsecured debt. – Mortgages.
Week 8	• **Sovereign default models**. – Methods. – Calibration. – Divide and conquer.
Week 9	• **Idiosyncratic and Aggregate Risk.** – "MIT shocks". – Transitions. – Krusell and Smith (1998). – Boppart, Krusell, and Mitman (forthcoming).
Week 10	• **Firms Heterogeneity in Macroeconomics.** – Cabral & Mata (2003), Midrigan & Xu (2014), and Pugsley, Sedlacek, Sterk (2017). – Hopenhayn (1992). Hopenhayn and Rogerson (1993). Gomes (2001).
Week 11	• **Firms, Finance, and Default.** – Hennessy and Whited (2007). – Gomes and Schmid (2010), Arellano, Bai, and Zhang (2012).
Week 12	• **Firms and Aggregate Fluctuations.** – Khan, Senga, and Thomas (2013). – Clementi and Palazzo (2016).
Week 13	• **Help with final projects at the time scheduled with each group.**
Week 14	• **Final projects presentation.**